Problem 1 – The many ways of writing momentum (30 Points)

Consider the heavy ball method below with time-varying or adaptive stepsizes:

$$w_{t+1} = w_t - \eta_t g_t + \beta_t(w_t - w_{t-1}), \quad w_0 = 0,\ \eta_0 = 0,$$

for all $t = 1, \ldots, T$. Show that this method can be written equivalently as

$$m_t = \hat{\beta}_t m_{t-1} + g_t, \quad w_{t+1} = w_t - \hat{\eta}_t m_t, \quad m_0 = 0$$

and

$$v_t = v_{t-1} - \bar{\eta}_t g_t, \quad w_{t+1} = \frac{\bar{\beta}_{t+1}}{1 + \bar{\beta}_{t+1}} w_t + \frac{1}{1 + \bar{\beta}_{t+1}} v_t, \quad v_0 = 0,$$

for specific parameters $\hat{\eta}_t$, $\hat{\beta}_t$, $\bar{\eta}_t$, $\bar{\beta}_t$.

Problem 2 – Recursion of De-biased SGD with Momentum (40 Points)

The goal of this exercise is to formally derive the recursive inequality of eq. 15 for De-biased SGD with Momentum in Slides on Variance Reduction. You may assume our standard SFO assumption and find it useful to apply Young's inequalities

$$\langle a, b\rangle \leq \frac{\|a\|^2}{2\lambda} + \frac{\lambda\|b\|^2}{2}, \qquad \|a+b\|^2 \leq (1+\lambda)\|a\|^2 + (1+\lambda^{-1})\|b\|^2$$

for suitable values of $\lambda > 0$.

Using the update of De-biased SGD with Momentum in eq. 14 in Slides on Variance Reduction, the definition of gradient estimation error, the L-Lipschitzness of $\nabla\ell(w, .)$ show that

$$\mathbb{E}_{z_t}[\|e_{t+1}\|^2|w_t] \leq (1-\beta)^2\|e_t\|^2 + 2\beta^2\sigma^2 + 2(1-\beta)^2L^2\eta^2\,\mathbb{E}\,\|m_t\|^2.$$

(Hint: You may need adding and subtracting suitable terms and expand the resultant quadratic expression with three terms, i.e. $\|a+b+c\|^2$. Then, you need to handle 3 square terms and 3 cross terms)

Problem 3 – Recursion of EF-SGD (30 Points)

The goal of this exercise is to formally derive the recursive inequality of eq. 13 for EF-SGD in Slides on Compression. You may assume our standard SFO assumption and find it useful to apply Young's inequalities

$$\langle a, b\rangle \leq \frac{\|a\|^2}{2\lambda} + \frac{\lambda\|b\|^2}{2}, \qquad \|a+b\|^2 \leq (1+\lambda)\|a\|^2 + (1+\lambda^{-1})\|b\|^2$$

for suitable values of $\lambda > 0$.

Using the update of EF-SGD method, the definition of compression error, and the assumption that the compression mechanism is contractive show that

$$\mathbb{E}_{\mathcal{C},z_t}[\|e_{t+1}\|^2|w_t] \leq (1-\delta^2)\|e_t\|^2 + \frac{\|\nabla f(w_t)\|^2}{\delta} + \frac{\sigma^2}{\delta}$$